

HOLDEN AGDM, AB, Canada

WMO: 712270

Lat: 53.1850N

Lon: 112.2461W

Elev: 688

StdP: 93.33

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.9	-27.9	-34.4	0.2	-30.8	-31.3	0.2	-27.8	11.8	-4.5	10.5	-6.5	3.2	220	0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	28.6	17.2	26.7	16.5	25.0	15.7	19.0	26.1	17.8	24.4	16.8	22.8	4.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.8	22.0	15.4	11.9	20.5	14.3	11.0	19.4	57.1	26.0	53.0	24.5	49.6	22.9	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.6	8.5	DB	-36.0	32.0	3.1	2.1	-38.2	33.5	-40.0	34.7	-41.7	35.9	-43.9	37.4	(4)
(5)			WB	-36.1	21.3	3.1	1.3	-38.3	22.3	-40.1	23.1	-41.8	23.8	-44.0	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-11.6	-11.7	-5.4	3.3	10.2	14.4	17.0	15.6	10.7	3.8	-5.0	-11.3	(6)	
	DBStd	12.14	8.52	7.79	7.37	5.24	4.25	2.96	2.80	3.18	4.41	4.97	6.93	7.75	(7)	
	HDD10.0	3366	669	608	477	204	50	4	0	2	44	198	451	659	(8)	
	HDD18.3	5782	927	841	736	450	253	123	59	94	231	449	701	918	(9)	
	CDD10.0	658	0	0	0	4	56	135	217	175	64	7	0	0	(10)	
	CDD18.3	32	0	0	0	0	1	4	18	8	1	0	0	0	(11)	
	CDH23.3	779	0	0	0	1	52	98	308	224	94	2	0	0	(12)	
	CDH26.7	172	0	0	0	0	7	13	79	48	25	0	0	0	(13)	
(14) Wind	WSAvg	4.1	4.4	4.1	4.3	4.7	4.5	4.1	3.6	3.5	3.9	4.2	4.1	4.1	(14)	
(15) Precipitation	PrecAvg	383	17	11	16	32	40	64	73	58	31	18	12	14	(15)	
	PrecMax	503	59	26	35	69	67	113	157	110	82	31	34	33	(16)	
	PrecMin	193	3	2	4	5	5	15	23	10	4	2	1	3	(17)	
	PrecStd	72	13	7	9	17	15	27	32	23	20	10	7	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	5.2	12.3	22.0	27.6	28.2	31.6	30.5	30.0	22.6	13.0	6.4	(19)	
		MCWB	2.2	1.9	6.0	10.4	13.7	16.7	19.3	17.2	15.7	12.3	7.2	3.0	(20)	
	2%	DB	3.7	3.2	7.9	18.6	24.6	26.1	28.9	27.8	26.1	17.4	8.2	3.7	(21)	
		MCWB	0.8	0.7	3.8	9.0	12.5	15.6	18.6	16.9	14.2	9.8	4.2	1.2	(22)	
	5%	DB	2.2	1.5	5.4	15.1	22.2	24.0	26.6	25.9	22.8	14.4	5.5	1.6	(23)	
		MCWB	0.2	-0.3	2.3	7.0	11.7	14.8	17.8	16.6	13.2	7.9	2.4	-0.3	(24)	
	10%	DB	0.2	-0.3	3.4	12.5	19.7	21.9	24.6	23.7	19.4	11.7	3.4	-0.4	(25)	
		MCWB	-1.3	-1.7	0.8	5.6	10.4	13.7	16.8	15.8	11.8	6.4	1.0	-2.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.4	7.0	11.0	14.7	18.5	21.2	20.1	16.3	12.6	7.2	3.1	(27)	
		MCDB	4.8	4.8	11.5	21.3	24.4	26.0	28.0	25.9	26.7	21.1	12.3	5.8	(28)	
	2%	WB	1.4	1.0	4.2	9.2	13.3	16.9	19.6	18.4	14.9	10.5	4.6	1.3	(29)	
		MCDB	3.4	2.8	7.6	18.0	22.4	23.2	26.5	25.2	24.5	16.5	7.7	3.5	(30)	
	5%	WB	0.2	-0.2	2.5	7.4	12.2	15.7	18.5	17.2	13.7	8.5	2.6	-0.2	(31)	
		MCDB	2.0	1.4	5.2	14.0	20.6	21.7	25.3	23.8	21.6	13.4	4.9	1.4	(32)	
	10%	WB	-1.3	-1.8	1.1	6.1	11.1	14.7	17.4	16.2	12.3	6.9	1.2	-2.0	(33)	
		MCDB	0.5	-0.2	3.0	11.7	18.3	20.6	23.4	22.6	18.7	10.9	3.2	-0.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.7	10.3	11.9	13.7	12.2	12.7	13.3	13.4	11.8	9.5	9.6	(35)	
		MCDBR	10.2	11.8	11.6	17.3	17.9	16.0	16.7	17.7	19.6	16.5	11.6	10.8	(36)	
	5% WB	MCWBR	8.4	9.9	8.1	9.1	8.0	7.2	7.9	8.6	9.6	9.4	8.0	8.8	(37)	
		MCWBR	10.3	11.4	11.0	15.9	16.1	13.4	15.2	15.6	18.0	15.7	11.0	10.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.258	0.264	0.304	0.331	0.348	0.365	0.359	0.372	0.312	0.279	0.253	0.245	(40)	
		taud	2.301	2.357	2.343	2.361	2.367	2.348	2.389	2.329	2.472	2.539	2.446	2.290	(41)	
	Ebn at Noon		736	861	886	903	901	885	884	849	872	840	757	682	(42)	
		Edh at Noon	58	76	96	108	113	117	110	111	84	63	50	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.01	2.06	3.57	4.69	5.76	6.03	6.23	5.11	3.55	2.07	1.01	0.75	(44)	
	RadStd		0.11	0.16	0.27	0.44	0.40	0.40	0.31	0.36	0.37	0.25	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (85 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page